

IPL Performance Analytics

IPL Data Analysis Project — 2008 to 2024

Objective: analyze match outcomes, team performance, player performance, batting, bowling, toss decisions, venues and season trends using Python, Pandas, NumPy, Matplotlib, Seaborn and Power BI.

Metric	Value
Matches	1,095
Seasons	17
Deliveries	260,920
Total Runs	347,756
Wicket Events	12,950
Normalized Venues	37

Data Cleaning & Preparation

Season labels were normalized to calendar years. Historical franchise names and venue spellings were standardized. Missing match-result rows were excluded from toss-win percentage calculations. The analytical tables were structured for reproducible Python visualization and Power BI import.

Season-level analysis

Season	Matches	Runs	Wickets	Avg Runs/Match
2008.0	58.0	17,937.0	690.0	309.26
2009.0	57.0	16,353.0	698.0	286.89
2010.0	60.0	18,883.0	725.0	314.72
2011.0	73.0	21,154.0	813.0	289.78
2012.0	74.0	22,453.0	858.0	303.42
2013.0	76.0	22,602.0	912.0	297.39
2014.0	60.0	18,931.0	674.0	315.52
2015.0	59.0	18,353.0	691.0	311.07
2016.0	60.0	18,862.0	666.0	314.37
2017.0	59.0	18,786.0	711.0	318.41
2018.0	60.0	19,901.0	722.0	331.68
2019.0	60.0	19,434.0	685.0	323.90
2020.0	60.0	19,416.0	677.0	323.60
2021.0	60.0	18,637.0	717.0	310.62
2022.0	74.0	24,395.0	912.0	329.66
2023.0	74.0	25,688.0	916.0	347.14
2024.0	71.0	25,971.0	883.0	365.79

Key Player Analysis

Top Batsmen	Runs	Top Bowlers	Wickets
V Kohli	8014	YS Chahal	213
S Dhawan	6769	DJ Bravo	207
RG Sharma	6630	PP Chawla	201
DA Warner	6567	SP Narine	200
SK Raina	5536	R Ashwin	198
MS Dhoni	5243	B Kumar	195
AB de Villiers	5181	SL Malinga	188
CH Gayle	4997	A Mishra	183
RV Uthappa	4954	JJ Bumrah	182
KD Karthik	4843	RA Jadeja	169

Toss Analysis

Across 1,095 matches, 704 toss winners chose to field (64.29%) and 391 chose to bat (35.71%). In the cleaned analysis, toss winners choosing to field won 53.9% of valid matches, compared with 45.4% when choosing to bat. Overall, the toss winner also won 554 of 1,090 matches (50.83%), indicating only a modest overall toss advantage.

Business Insights

- Mumbai Indians lead the source analysis in match wins with 144, followed by Chennai Super Kings (138) and Kolkata Knight Riders (131).
- Virat Kohli is the leading run scorer with 8,014 runs in the 2008–2024 delivery data.
- Yuzvendra Chahal leads wicket events with 213 in the analytical dataset.
- AB de Villiers has the most Player of the Match awards with 25.
- 2024 has the highest total runs (25,971) and highest average runs per match (365.79).
- Run rate increased from 7.21 in 2009 to 9.11 in 2024, indicating a strong long-term scoring-intensity increase.
- Wankhede Stadium is the strongest venue in the Mumbai Indians venue-win analysis with 42 wins.
- The 2022–2024 seasons combine expanded schedules with higher scoring totals.

Python Visualizations

Figure 1

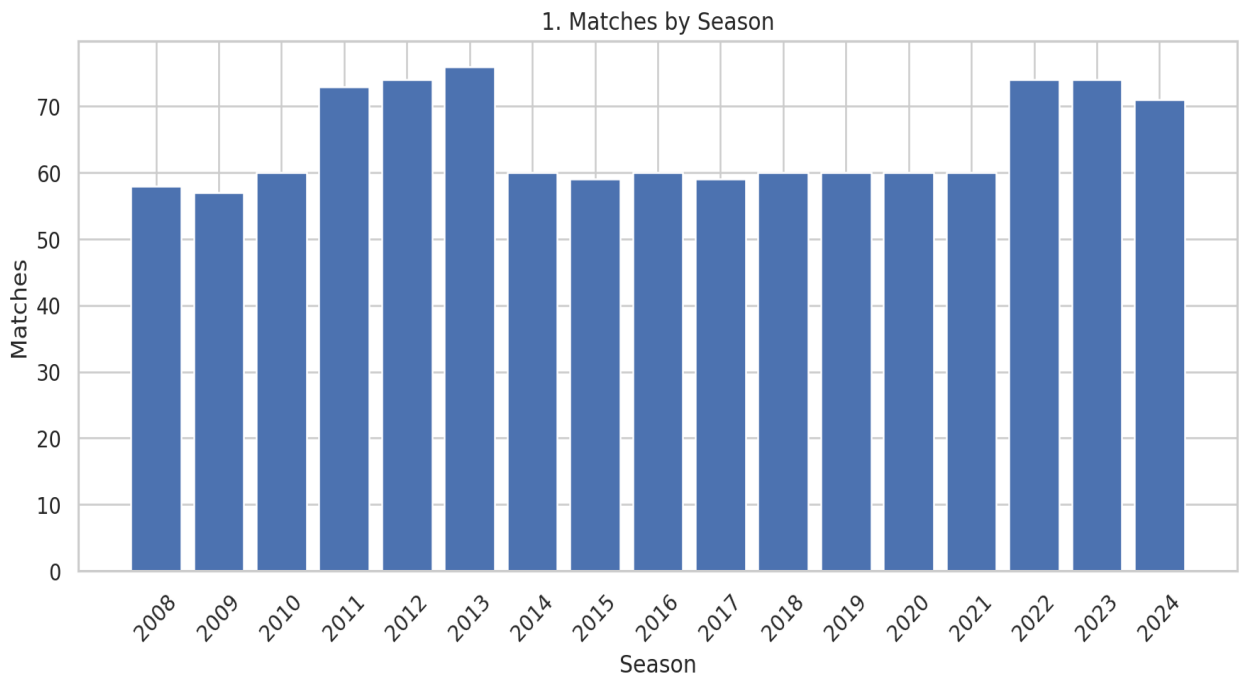


Figure 2

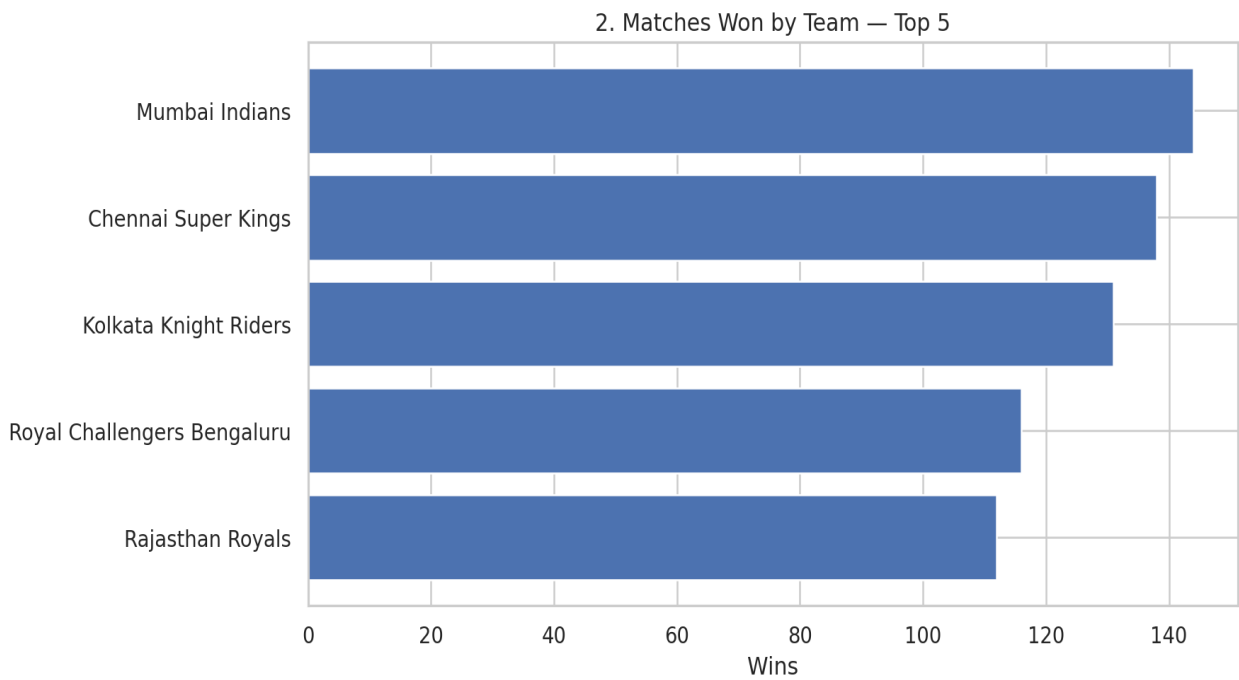


Figure 3

3. Top 10 Run Scorers

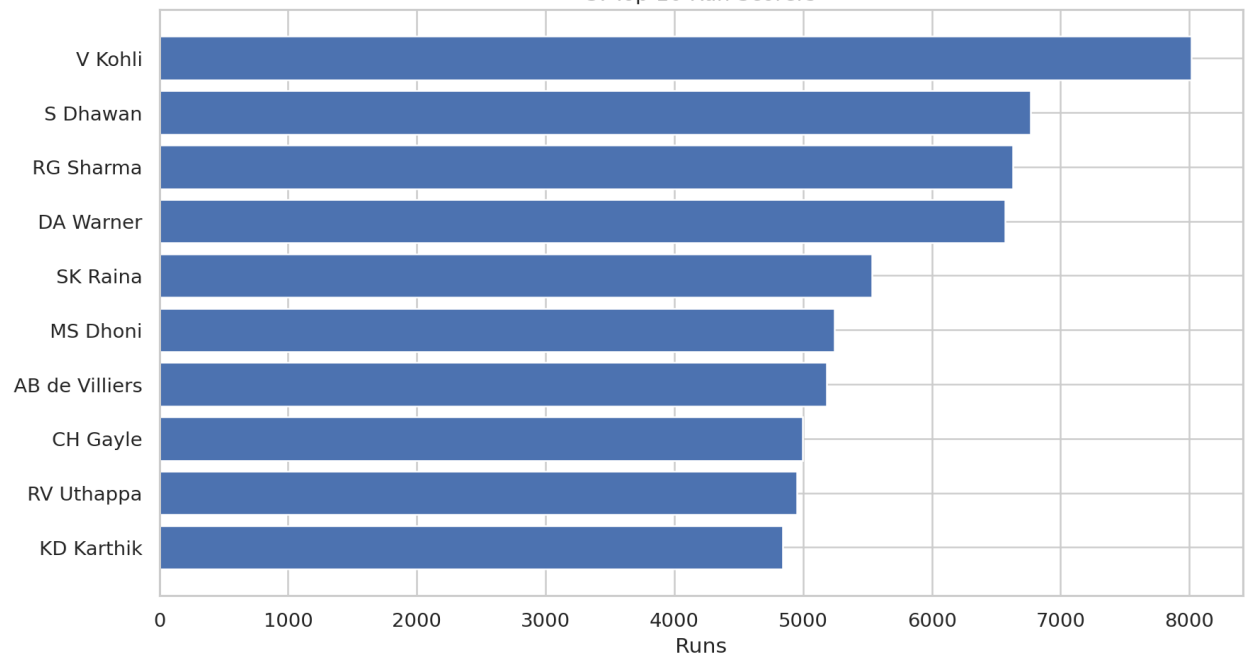


Figure 4

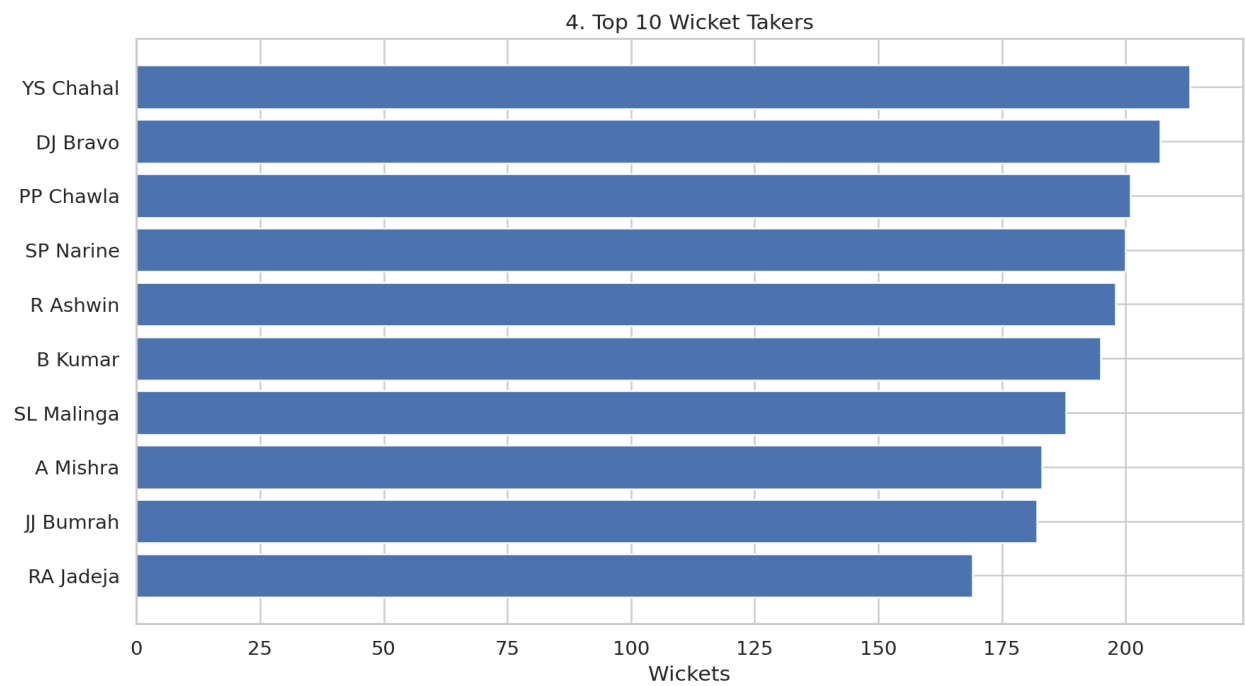


Figure 5

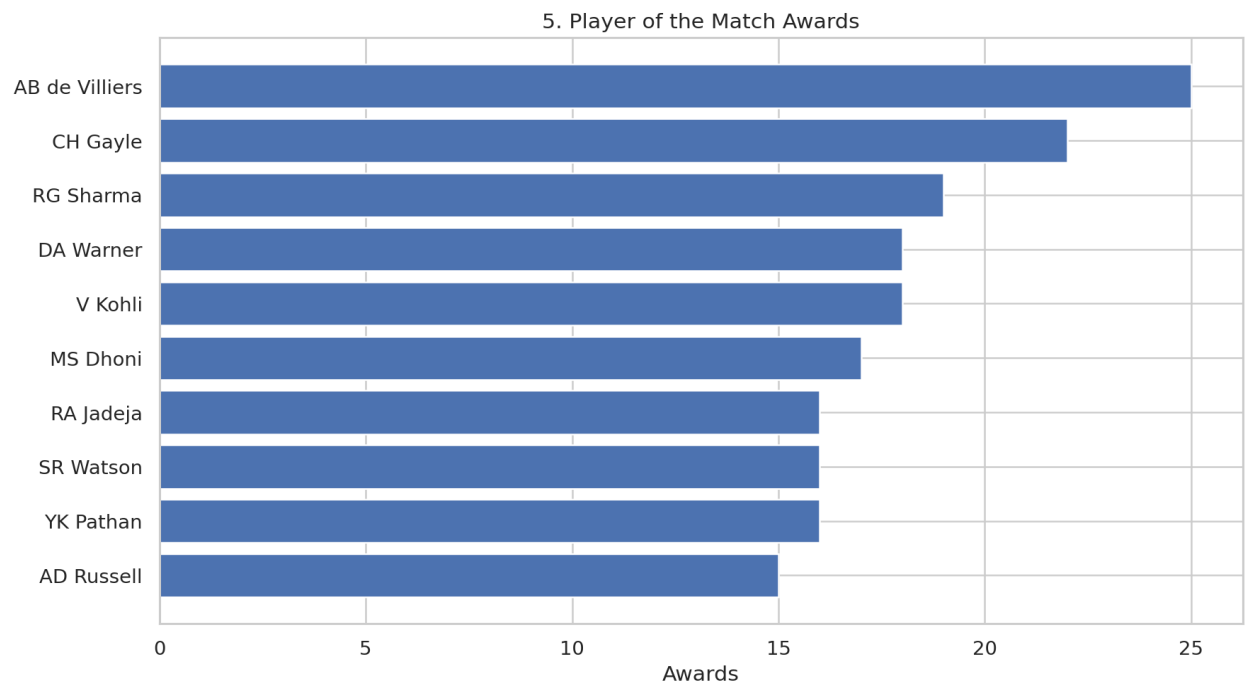


Figure 6

6. Toss Decision Distribution

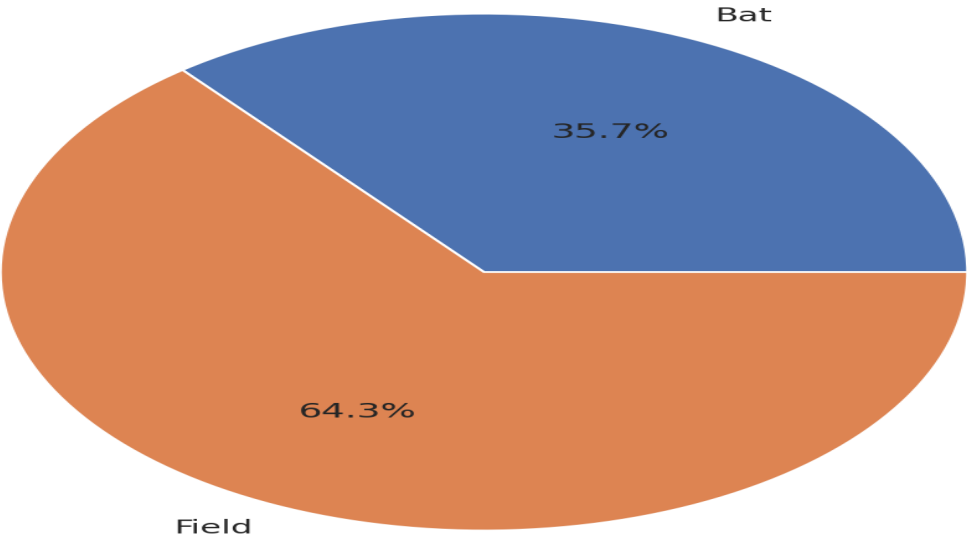


Figure 7

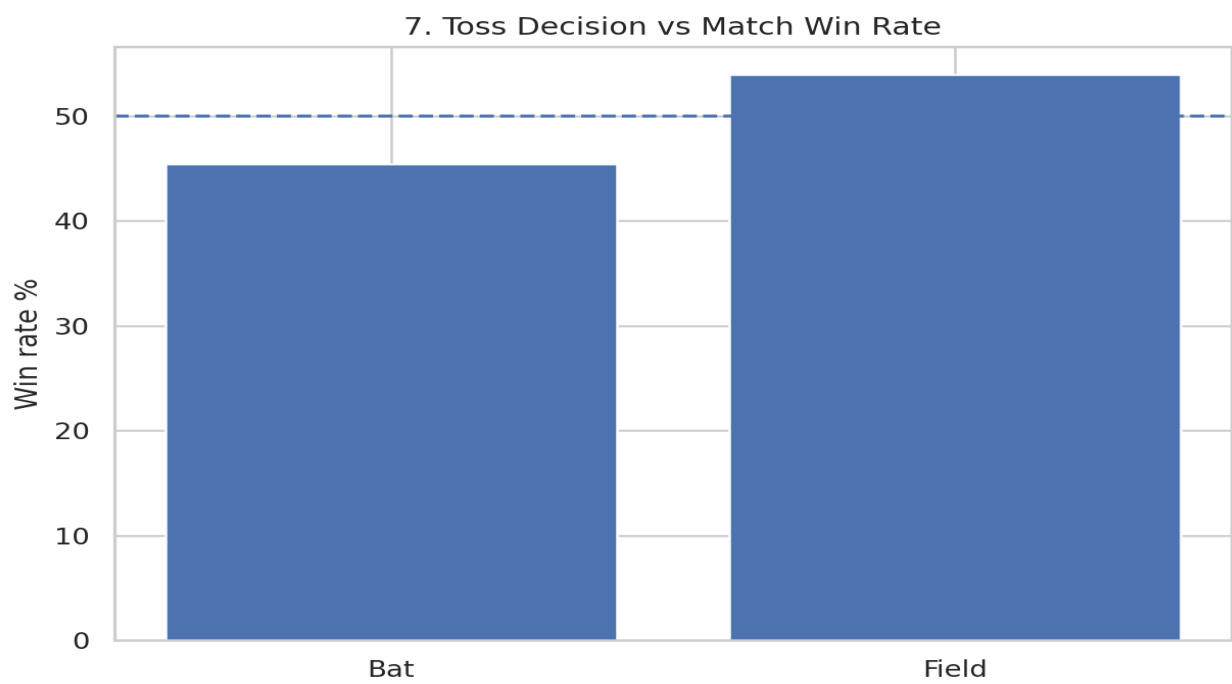


Figure 8

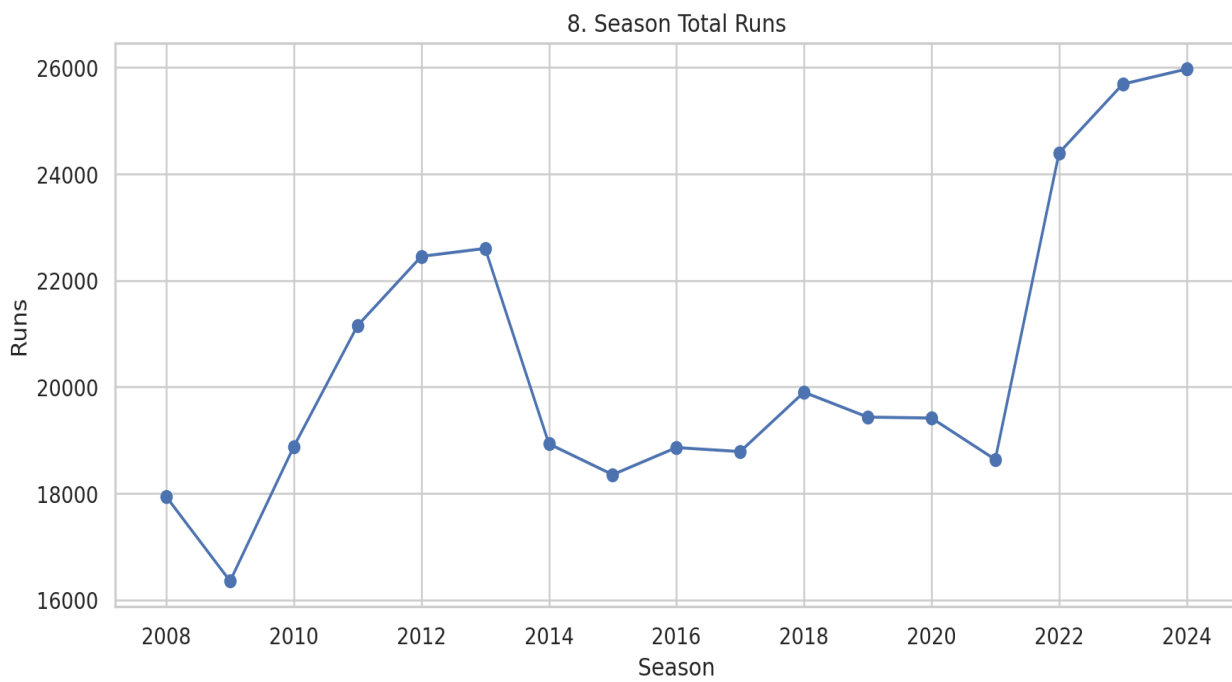


Figure 9

9. Season Total Wickets

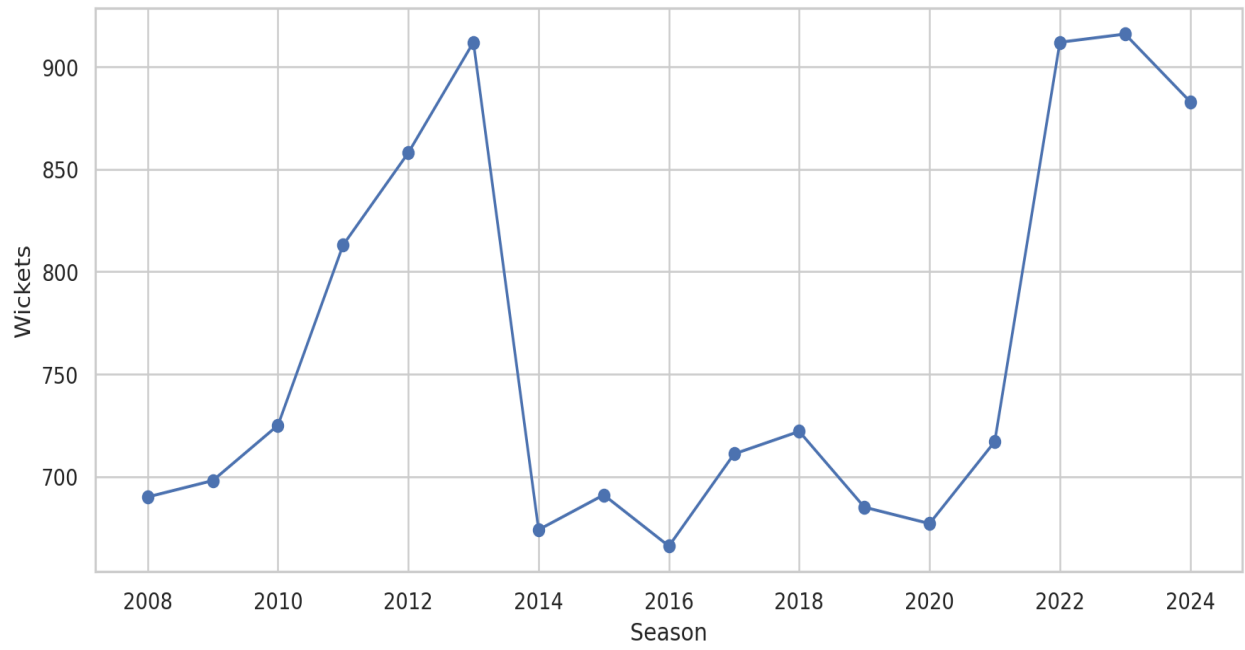


Figure 10

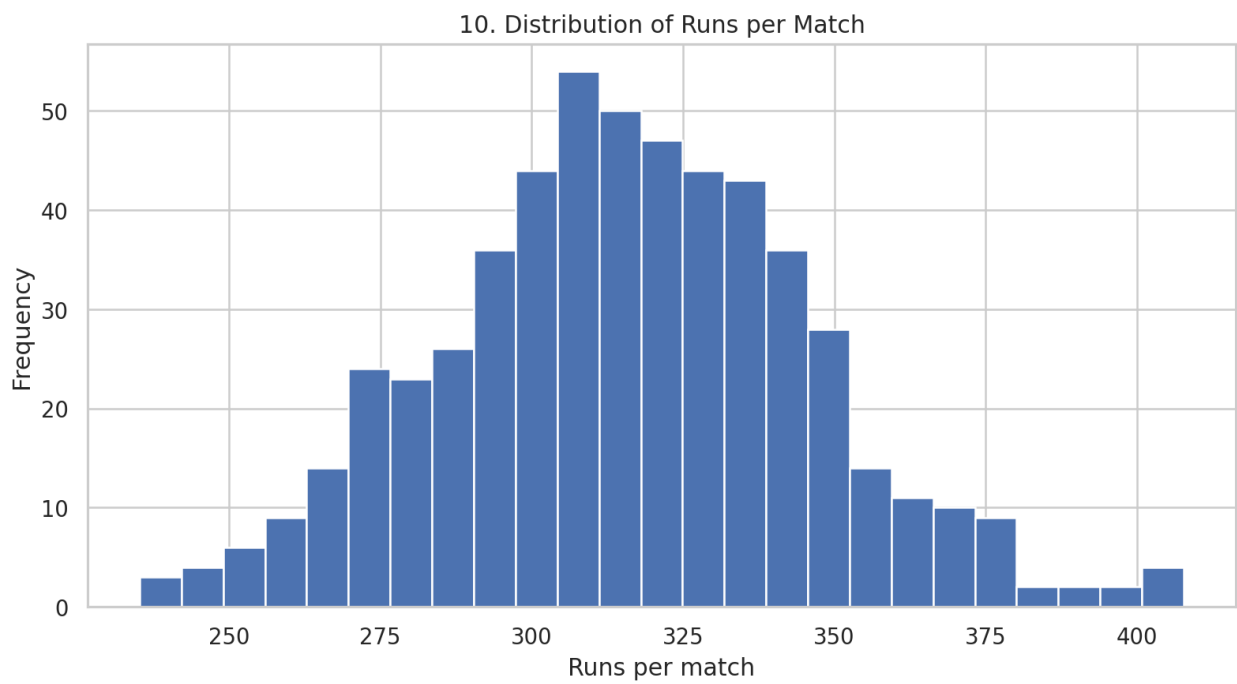


Figure 11

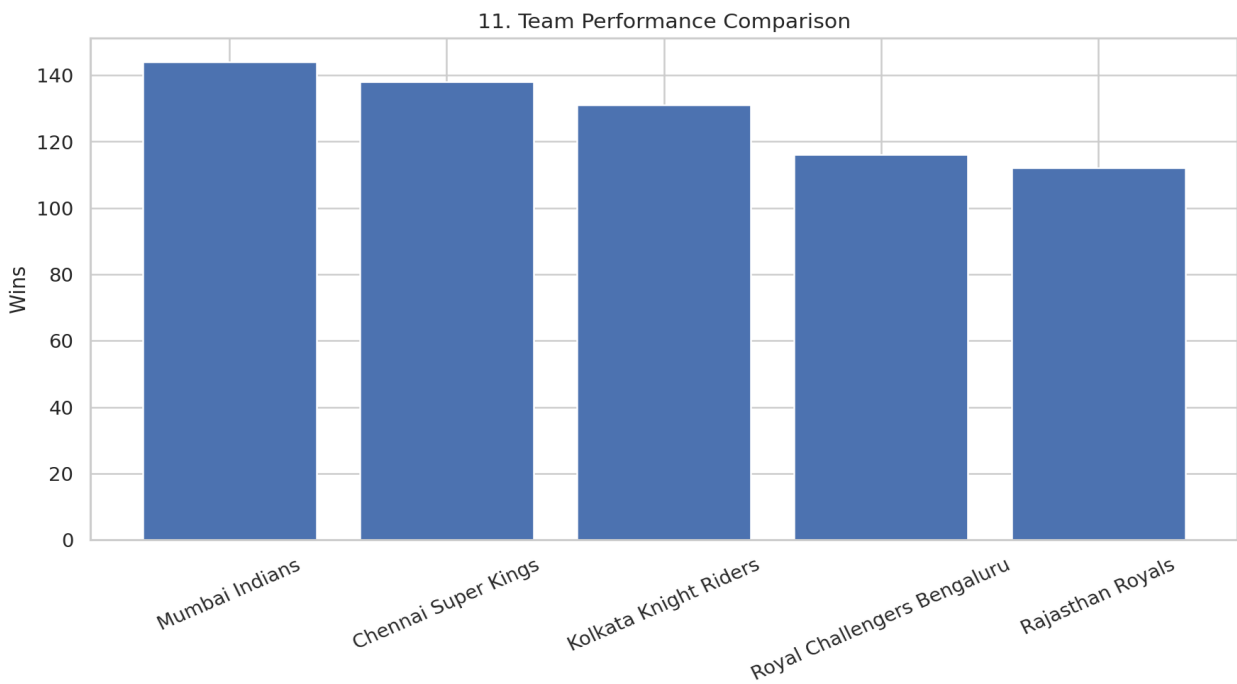


Figure 12

12. Venue Analysis — Selected High-Use/Successful Venues

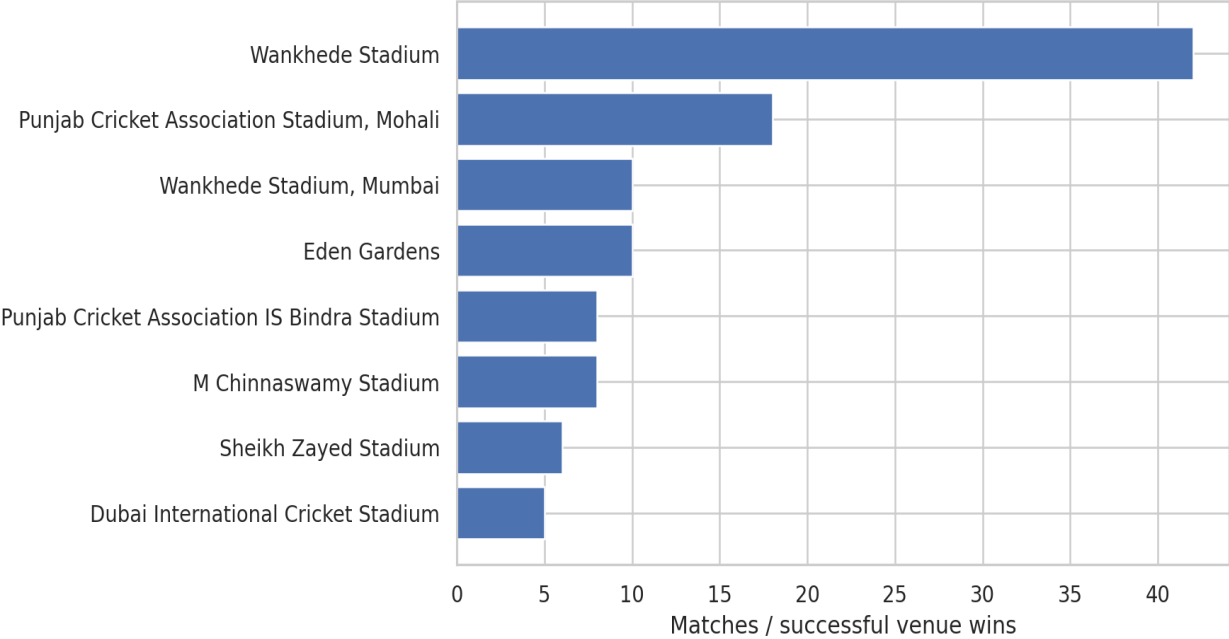


Figure 13

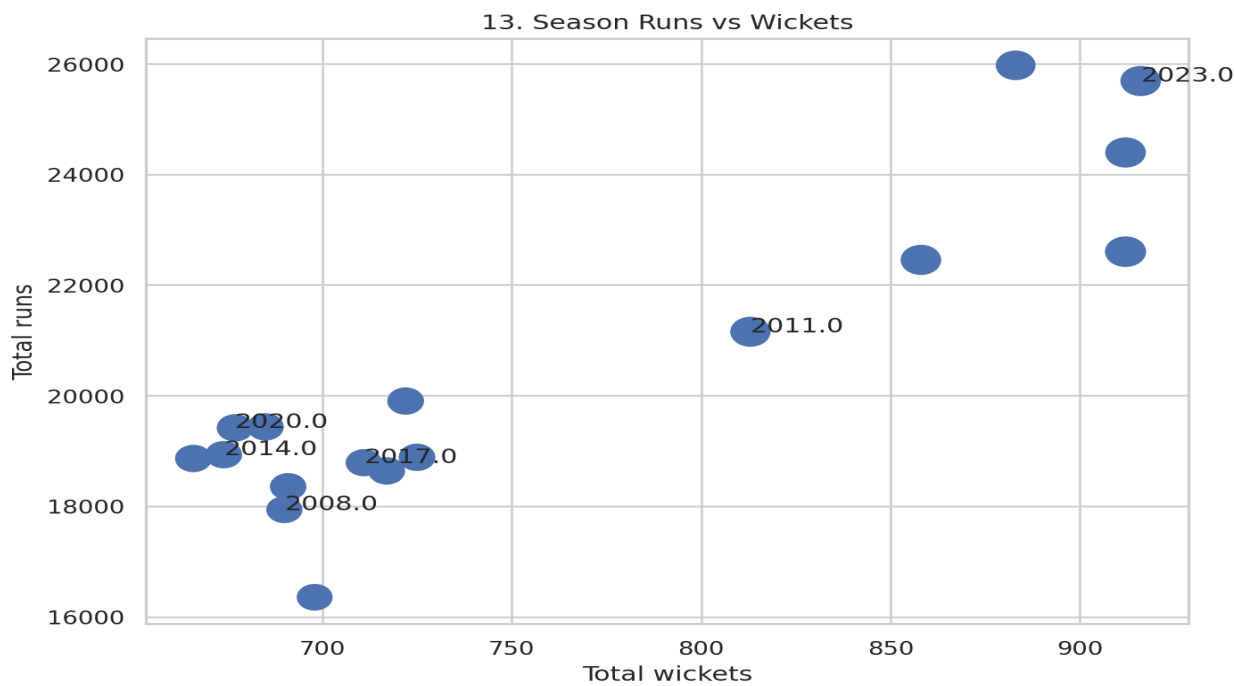
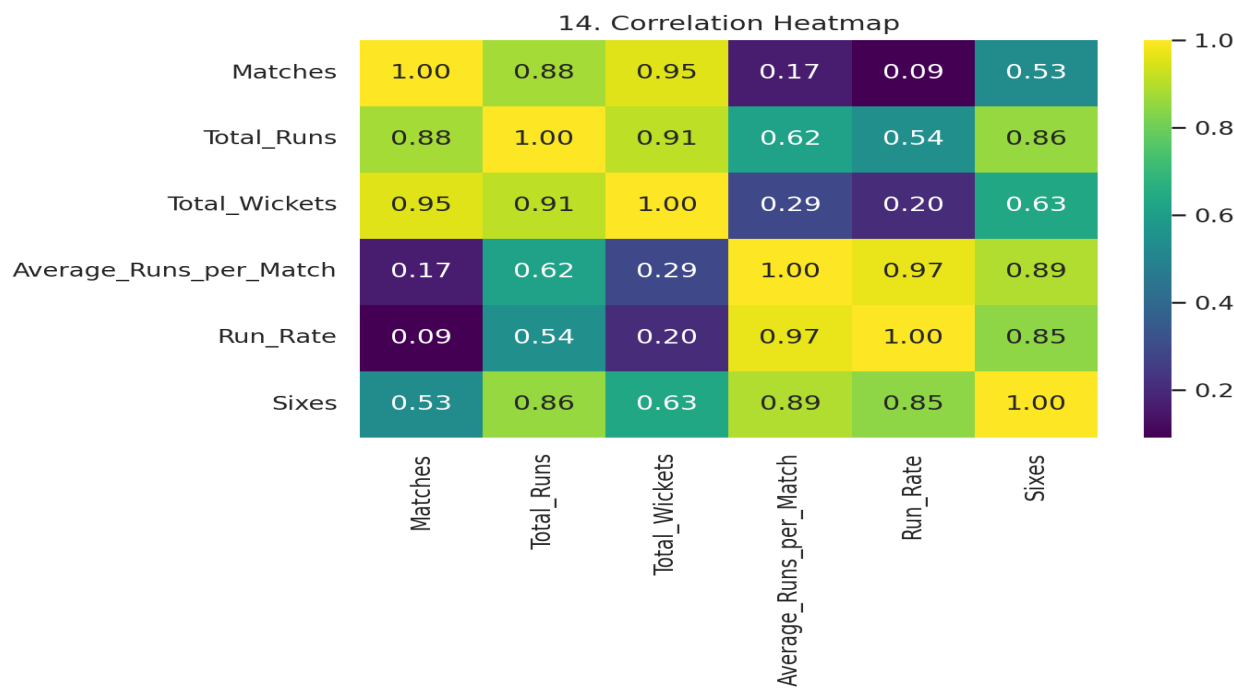


Figure 14



Power BI Dashboard Specification

Dashboard title: IPL Performance Analytics Dashboard. Recommended KPI cards: Total Matches, Total Runs, Total Wickets, Total Teams, Total Players and Average Runs per Match. Recommended visuals: matches by season, team wins, top 10 batsmen, top 10 bowlers, runs by season, venue analysis, toss decision donut, Player of Match bar and a team performance table/matrix. Add Season, Team and Venue slicers. The native PBIX must be saved from Power BI Desktop after importing the supplied CSVs and creating the supplied DAX measures.

Conclusion

The analysis shows a clear increase in IPL scoring intensity over time, strong long-term performance by established franchises, sustained dominance by a small group of elite batters and bowlers, and only a modest overall advantage from winning the toss. Power BI can convert these findings into an interactive dashboard for business-style exploration.